

Pleural metastatic disease from glioblastoma multiforme.

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Glioblastoma multiforme (GBM) is a malignant primary brain neoplasm with poor survival. Extracranial GBM dissemination is very uncommon, but is increasingly being recognized. The imaging features of metastatic pleural GBM are not specific for a primary central nervous system neoplasm. In this case, extracranial GBM metastases presented as circumscribed posterobasal pleural nodules. Diagnoses may be established with percutaneous or thoracoscopic biopsy. Radiologists should be familiar with the possibility of the extracranial spread of GBM because as therapeutic improvements provide increased local control of the primary tumor, improving patient survival, extracranial spread of disease will be increasingly encountered on thoracoabdominal imaging studies.

A Rare Case of Metastases from a High-grade Astrocytoma to the Pleura, Bones, and Liver within Six Months of Diagnosis.

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High grade astrocytomas such as anaplastic astrocytoma and glioblastoma multiforme are aggressive central nervous system malignancies with a poor prognosis. Due to shortened survival times, their devastating effects are usually localized intracranially and rarely metastasize outside of the central nervous system. When metastases occur, they usually present in patients with longer survival times and they typically coincide with a primary site recurrence. We present a rare case of metastases from a high-grade astrocytoma/glioblastoma to the pleura, bones and liver within six months of diagnosis, without primary site recurrence.

Abnormal Schwannoma-like Growth of multiple, multifocal BRAF V600E-positive Glioblastoma in the Interior Acoustic Canal with Leptomeningeal Infiltration: a case report.

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Glioblastoma belongs to the most common and most aggressive tumor entity of the central nervous system with a poor prognosis of only few months. Once manifested, it grows fast and diffusely by infiltrating the surrounding brain parenchyma. Despite its aggressive behavior, glioblastoma rarely presents with multiple lesions and metastasis to intra- and extracranial tissues. Therefore, metastasized, multiple glioblastoma is limited to case reports. Our case describes an atypical primary bilateral manifestation of BRAF V600E-positive epithelioid glioblastoma with rapid metastasis and meningeosis glioblastoma while under adjuvant chemoradiotherapy. A 60-year-old Caucasian male patient presented with a seizure and numbness in his left arm. He was diagnosed with an abnormal primary bilateral manifestation of multiple, multifocal BRAF V600E-positive and isocitrate dehydrogenase (IDH) wild-type intracranial epithelioid glioblastoma with O6-methylguanine-DNA methyltransferase methylation (MGMT) at 12%. While being under the adjuvant chemoradiotherapy with temozolomide, the patient developed left-sided facial nerve weakness and hearing loss, dysarthria, and severe gait instability. Cranial magnetic resonance imaging showed that glioblastoma lesions advanced rapidly with a schwannoma-like growth pattern by invading the left internal acoustic meatus, adjacent cranial nerves,

and leptomeninges. A lumbar puncture confirmed meningeosis glioblastoma. Four months after the initial diagnosis of glioblastoma, the patient died from the complications of the fast and diffuse metastasis. Glioblastoma rarely presents with metastases despite its aggressive and rapidly growing nature. Our case should increase awareness of symptom tracking in patients with glioblastoma to intervene early and efficiently. Moreover, refractory therapies for glioblastoma should underline the importance of personalized medicine.

Postoperative Extra-Cranial Metastasis of Glioblastoma: A Case Report.

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Glioblastoma (WHO grade IV) is the most common malignant tumor of neural tissues in adults as a primary tumor. Because of blood brain barrier and short median survival of patients with glioblastoma, metastasis of this tumor is very rare. A 46-year-old man was admitted to Sina hospital with chief complaint of headache and visual impairment. After neuro-radiologic evaluation the patient underwent surgery. Pathologic examination of the tumor confirmed the diagnosis of glioblastoma multiforme. Cytogenetic study of the tumor cells confirmed GBM IDH1 wild type with TERT mutation and EGFR amplification. Two months after surgical resection, the tumor recurred with involvement of the dura matter. After the second operation, metastasis to the pelvic cavity and cervical lymph node was found. Almost all cases of glioblastoma metastasis had undergone surgery or any manipulation; this fact suggests that iatrogenic intra-vascular seeding of tumor cells at the time of resection and disruption of blood brain barrier could cause extra-neural metastasis.

Subcutaneous tissue metastasis from glioblastoma multiforme: A case report and review of the literature.

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Glioblastoma multiforme is the most common primary brain tumor, despite an aggressive clinical course, less than 2% of patients develop extraneural metastasis. We present a 72-year-old male diagnosed with a right temporal glioblastoma due to headache. He underwent total gross resection surgery and after that the patient was treated with adjuvant therapy. Five months after the patient returned with trigeminal neuralgia, and MRI showed an infratemporal cranial mass which infiltrates masticator space, the surrounding bone, the temporal muscle and superior cervical and parotid lymph nodes. The patient underwent a new surgery reaching partial resection of the temporal lesion. After that the patient continued suffering from disabling trigeminal neuralgia, that's why because of the bad clinical situation and the treatment failure we decided to restrict therapeutic efforts. The patient died 3 weeks after the diagnosis of extracranial metastasis.

Extracranial metastasis of glioblastoma: A case report and literature review.

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Glioblastoma (GBM) is the most common malignant tumor of the central nervous system. Extracranial

metastasis is rare, accounting for only 0.4 %-0.5 % of all GBM patients. The pathways and mechanisms involved are still unclear. We reported a rare case of GBM with multiple bone metastases, highly suspected of abdominal metastasis. This 20 year old woman underwent surgery in March 2017 and underwent postoperative radiotherapy and chemotherapy. In July 2018, she underwent a second surgery due to intracranial recurrence and also underwent radiotherapy and chemotherapy after the surgery. She experienced pain in the lumbosacral region in May 2019, abdominal magnetic resonance imaging (MRI) showed metastases to the ilium, sacrum, and multiple lumbar vertebrae. In August 2019, a lump was discovered at the sternum and biopsy was performed, pathological examination confirmed it as GBM. During this period, the patient's condition was briefly controlled after receiving palliative radiotherapy, chemotherapy, and targeted treatment. Surprisingly, the patient later developed highly suspected malignant ascites, and further anti-tumor treatment was refused. She died 7 months after diagnosis of extracranial metastases. This patient with GBM had multiple bone metastases and highly suspected abdominal metastasis after two operations. Chemotherapy, radiotherapy and Targeted therapy extend the survival period and improve the quality of life. We believe that the patient's extracranial metastases may have occurred through blood. Young "long-term survivors" who have undergone surgery seem to have a higher risk of extracranial metastasis. Timely detection and early treatment can improve the overall quality of life of the patient.

Extracranial Metastases From Glioblastoma With Primitive Neuronal Components on FDG PET/CT.

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A 17-year-old girl with World Health Organization grade IV glioblastoma with primitive neuronal components (histone H3 G34-mutant and IDH1 wild type) underwent whole-body FDG PET/CT staging due to vertebral metastases on initial MRI. PET/CT revealed extracranial metastatic disease with spinal leptomeningeal dissemination, osseous metastases, and peritoneal seeding via a ventriculoperitoneal shunt. Glioblastoma is uncommon in pediatric patients and particularly those with primitive neuronal components. Extracranial metastases from glioblastoma are more common in those with primitive neuronal components. This case demonstrates the utility of FDG PET/CT for revealing distant metastases from glioblastoma.

Spinal and cervical nodal metastases in a patient with glioblastoma.

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This article presents the rare case of a 54-year-old gentleman with primary glioblastoma developing multiple extracranial metastases 7 months after diagnosis. Initially, the patient complained of progressive headaches, confusion, and weakness of the left arm. Magnetic resonance imaging of the brain showed a right temporoparietal tumor with substantial surrounding subcortical edema and midline shift to the left. Two consecutive craniotomies resulted in complete microsurgical resection of the lesion. Histology was consistent with a World Health Organization grade IV, IDH-wildtype glioblastoma. Further treatment was standard chemoradiation including intensity-modulated radiotherapy with oral temozolomide chemotherapy. Seven months after diagnosis, the cranial lesion progressed, and the patient developed painful metastases in multiple bones and suspicious right-sided cervical lymph nodes.

Immunohistochemistry and molecular signature supported the case of a metastatic glioblastoma. Further treatment was palliative radiotherapy of the spinal lesions along with symptomatic pain management. Extracranial metastasis of glioblastoma is a rare complication of which only a few cases have been reported in the literature. Little is known about the precise mechanisms of tumor dissemination and the appropriate treatment.

Increasing C-reactive protein levels in a patient with glioblastoma with lymph node metastasis: a case report.

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Glioblastoma usually recurs locally and extracranial metastases are rare. Most patients with extracranial metastases experience recurrence of the primary intracranial tumor. Lymph node metastases are often detected based on lymphadenopathy or symptoms caused by other metastatic sites. Herein, we report a case of glioblastoma with lymph node metastasis in which the patient was asymptomatic but exhibited gradually increasing C-reactive protein levels prior to becoming febrile 9 months after the initial C-reactive protein increase. Diagnosis of lymph node metastasis that was delayed because the patient had a fever of unknown origin, no signs of infection, and the primary intracranial tumor did not recur. Chest computed tomography indicated supraclavicular, mediastinal, and hilar lymphadenopathy, and biopsy identified lymph node metastasis of glioblastoma. This is the fifth reported case of lymph node metastasis without intracranial recurrence. C-reactive protein levels may be a diagnostic marker for lymph node metastasis in patients with glioblastoma. Further evaluation is needed to elucidate the role of CRP in glioblastoma with lymph node metastasis.

Intracerebral metastasis of glioblastoma multiforme. Case report and literature review.

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Glioblastoma multiforme is the most common primary malignant brain tumor in adults. It's characterized by a high malignancy and rapid, frequent tendency to local recurrence. Distant metastases disseminated in the brain compared to the primary lesion and outside the central nervous system are rarely reported in the literature. The case which is being presented is of a 53 year old man operated in 2008 because of Glioblastoma multiforme IV WHO in the left periventricular parietal region, in which the main symptoms were the Gerstmann syndrome, mixed aphasia and memory disturbance. The patient was operated totally followed by adjuvant radiotherapy and chemotherapy. Two years later epileptic seizures and aphasia were intensified. Due to this adverse symptoms MRI was ordered, which revealed a tumor in the left periventricular temporal region in different location compared to the primary lesion. The patient was operated by temporal lobectomy. Histopathological diagnosis was Glioblastoma multiforme IV WHO. According to the literature we analyzed the natural GBM tumor features and factors responsible for possibility to appear of the same type of tumor in another location of the brain.